

INTERNATIONAL GCSE **CHEMISTRY**

9202/2

Paper 2

Mark scheme

November 2023

Version: 1.0 Final



2 3 B Y 9 2 0 2 / 2 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(...) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

'Ignore' is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

'Do **not** accept' means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	covalent		1	AO1 3.2.1g 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	3		1	AO1 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	graphite is slippery		1	AO2 3.2.3c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	(graphite has) delocalised electrons	allow (graphite has) free electrons	1	AO1 3.2.3d
	(which) can move through the structure / graphite	ignore throughout	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	any two from: • drug delivery in medicine • lubricants • catalysts • reinforcing materials in sports equipment	allow reinforcing materials in a named sports equipment (eg tennis rackets)	2	AO1 3.2.3e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	1–100 nm		1	AO1 3.2.4

Total Question 1		8
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	type of metal		1	AO4 3.3.1.1b 3.3.1.1h

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	(mean =)			AO2 3.3.1.1b 3.3.1.1h
	$\frac{5.3 + 5.7 + 5.6}{3}$		1	
	= 5.533333...		1	
	= 5.5 (°C)	allow an answer correctly rounded to 2 significant figures from an incorrect calculation which uses values from the question	1	

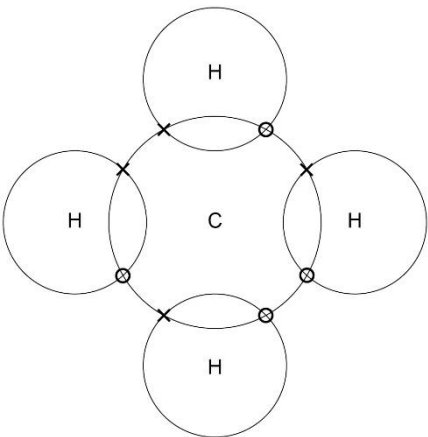
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	(metal) magnesium	allow Mg	1	AO3 3.3.1.1b
	(reason) has the highest temperature rise	allow ecf from Question 02.2	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	add insulation to the boiling tube		1	AO4 3.3.1.1b

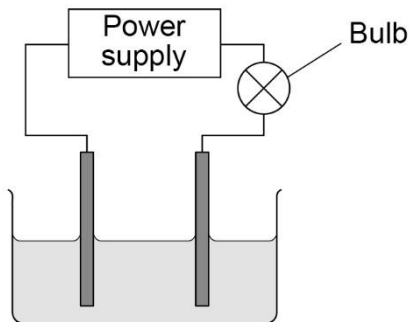
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	magnesium sulfate	allow MgSO_4	1	AO2 3.3.1.1h
	copper	allow Cu	1	

Total Question 2		9
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	4 shared pairs of electrons with no non-bonded electrons on hydrogen and / or carbon atoms	allow any combination of dots, crosses, circles or e^{-} for electrons allow 1 mark for a pair of electrons in an overlap an answer of  scores 2 marks	2	AO1 3.2.1g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	(substance) B (reason) simple molecules do not conduct electricity	MP2 is dependent upon MP1 being awarded	1 1	AO3 3.2.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3			1	AO4 3.2.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	increases		1	AO3 3.10.1.2c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	weak intermolecular forces	allow weak forces of attraction between molecules	1	AO2 3.2.2d
	(which) are easily overcome	allow (which) do not take much energy to overcome	1	

Total Question 3	8
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	bubbles of gas magnesium becomes smaller	allow bubbling / fizzing	1	AO1 3.3.1.1a
		allow magnesium disappears	1	
		allow becomes warmer		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	any two from: - length of magnesium - volume of hydrochloric acid - concentration of hydrochloric acid		2	AO4 3.8.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	175 (s)	allow a value in the range 170–175 (s)	1	AO3 3.8.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	(hydrochloric) acid is used up		1	AO2 3.8.1c

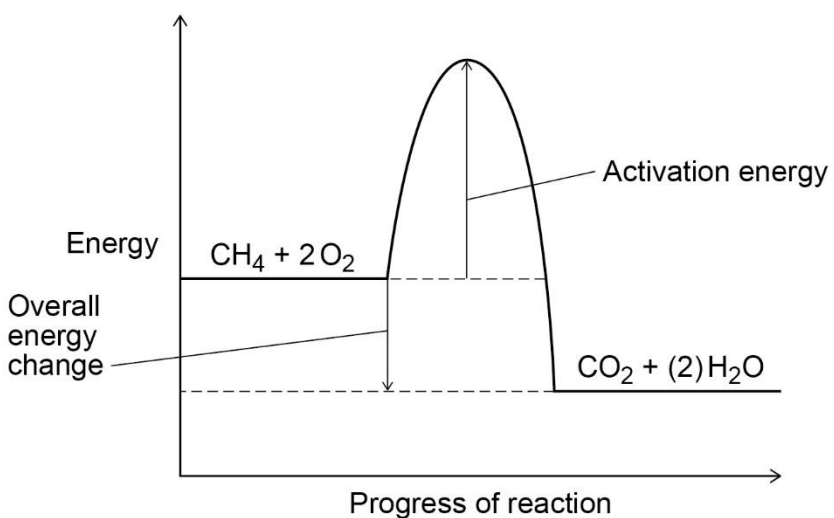
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	(volume of gas =) 29 cm ³	allow a value in the range 28–30 cm ³	1	AO2 × 3 AO3 × 1 3.8.1a
	$\left(\text{rate} = \frac{29 - 0}{50 - 0}\right) = \frac{29}{50}$	allow correct use of an incorrectly determined volume	1	
	= 0.58		1	
	(unit =) cm ³ / s		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	a steeper line to the left of the original line starting at 0,0		1	AO3 3.8.1c
	levels off at 80 cm ³ of gas	allow a tolerance of $\pm \frac{1}{2}$ a small square	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.7	the rate of reaction increases		1	AO1 3.8.1c
	(because) the particles move faster	allow (because) the particles have more energy	1	
	(so there are) more frequent collisions	allow (so) more likelihood of collisions	1	

Total Question 4		15
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	correct shape for exothermic reaction		1	AO2 3.9.2d
	labelled activation energy	allow correct label on a diagram drawn for an endothermic reaction	1	
	labelled overall energy change	allow correct label on a diagram drawn for an endothermic reaction	1	
	(product level) labelled $\text{CO}_2 + (2)\text{H}_2\text{O}$	allow carbon dioxide and water allow correct label on a diagram drawn for an endothermic reaction	1	
		an answer of		
				
		scores 4 marks		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	(bond breaking) $((2 \times 436) + 498 =) 1370$	allow correct use of incorrectly determined values in step 1 and/or step 2	1	AO2 3.9.2f
	(bond making) $(4 \times 464 =) 1856$		1	
	$(1370 - 1856 = -) 486$		1	
	(enthalpy change $\Delta H =$) $- 486$ (kJ / mol)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	hydrogen only releases water	allow methane also releases carbon dioxide	1	AO3 3.9.2f
	hydrogen is obtained from a renewable source	allow methane is obtained from a non-renewable source	1	

Total Question 5		10
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Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	as a solvent		1	AO1 3.10.3.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	compound X could be ethanol because ethanol is also flammable		1	AO3 3.10.3.1b
	there is no reaction with sodium so compound X is not ethanol	allow if compound X was ethanol there would be a reaction with sodium	1	
	compound X could be ethanol because ethanol also dissolves in water to form a neutral solution		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	oxidation		1	AO1 3.10.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C}=\text{O} \\ \quad \\ \text{H} \quad \text{O}-\text{H} \end{array} $	allow 1 mark for single bond from second C atom to O of OH and single bond between O and H allow 1 mark for double bond from second C atom to another O atom	2	AO1 3.10.3.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	ethyl ethanoate		1	AO1 3.10.3.3a

Question	Answers	Mark	AO/ Spec. Ref.
06.6	Level 3: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5–6	AO2 × 4 AO3 × 2 3.8.1.gh 3.8.2.bcde 3.10.1.2f
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3–4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	
	No relevant content	0	
	Indicative content: Yield 300 °C gives the highest percentage yield of ethanol of the three temperatures tested - lower temperature favours the exothermic / forward reaction 65 atmospheres leads to the steepest increases in percentage yield of ethanol at any temperature - high pressure favours the side with less moles of gas / RHS. Rate - higher temperature would increase rate of reaction - higher pressure would increase rate of reaction - using a catalyst increases rate of reaction. Other factors - using a catalyst reduces energy costs due to reduced activation energy - higher temperatures lead to increased energy costs - higher pressure leads to increased energy costs - higher pressure requires stronger reaction vessels, leading to increased costs. Compromise - chosen temperature is a compromise between rate and yield - chosen temperature is a compromise between rate and cost (of energy used) - chosen pressure is a compromise between rate and cost (of energy used) - chosen pressure is a compromise between yield and cost (of energy used).		

Total Question 6	14
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	H ⁺		1	AO1 3.5.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	burette		1	AO4 3.6.4b Required practical 3 titration

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	any two from: - swirl the flask - stand the flask on a white tile - add sulfuric acid dropwise as permanent colour change nears - repeat and calculate a mean	allow repeat until concordant results obtained allow repeat until results within 0.1 cm ³	2	AO4 3.6.4b Required practical 3 titration

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.4	(indicator) methyl orange (colour change) yellow to red		1	AO4 3.6.4c
	OR (indicator) phenolphthalein (1) (colour change) pink to colourless (1)		1	
		if no other marks awarded, allow 1 mark for universal indicator turns from purple / blue to green / yellow / orange / red		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	(moles sulfuric acid =) $\frac{13.5}{1000} \times 0.2$ = 0.00270 mol		1	AO2 3.6.4c
	(moles sodium hydroxide = 0.00270 $\times$ 2 =) 0.00540 mol		1	
	(concentration = $\frac{0.00540}{0.0250}$) = 0.216 (mol/dm ³)		1	
		allow correct use of incorrectly determined moles of sulfuric acid allow correct use of incorrectly determined moles of sodium hydroxide		

Total Question 7		10
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	copper chloride solution		1	AO2 3.3.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	so the ions are free to move		1	AO1 3.3.2a 3.3.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	copper ions are positively charged		1	AO2 3.3.2c
	(so are) attracted to the negative electrode	allow (so) move to the negative electrode	1	
	(where the ions) gain electrons	allow (where the ions) are reduced	1	
	to form copper atoms		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	(anomalous result) 12 minutes	allow 2.2 cm ³	1	AO3 × 1 AO4 × 1 3.3.2c
	(reason) read volume before 12 minutes	allow read volume too early	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.5	(volume of) gas collected increases slowly until 7 minutes	allow a value in the range 6–8 minutes	1	AO3 3.3.2c 3.3.2g
	(because) chlorine gas dissolves in water at the beginning so only some is collected		1	
	after 7 minutes the rate of increase is faster		1	
	(because) no more chlorine gas can dissolve in the water	if MP1 not awarded allow 1 mark for increasing rate of gas collection until 7 minutes if MP3 not awarded allow 1 mark for after 7 minutes rate of gas collection is constant	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.6	(volume collected at 16 minutes =) 4.8 (cm ³)		1	AO2 3.6.4d
	$\left(\text{conversion } \frac{4.8}{1000} =\right)$ 0.0048 (dm ³)	allow correct conversion of an incorrectly determined volume of gas	1	
	(moles of gas =) $\frac{0.0048}{24}$		1	
	= 0.00020 (moles)	allow 2.0×10^{-4} (moles)	1	
	alternative approach (volume collected between 16 and 20 minutes = 6.6 - 4.8 =) 1.8 (cm ³) (1)			
	$\left(\text{conversion } \frac{1.8}{1000} =\right)$ 0.0018 (dm ³) (1)	allow correct conversion of an incorrectly determined volume of gas		
	(moles of gas =) $\frac{0.0018}{24}$ (1)			
	= 0.000075 (moles) (1)	allow 7.5×10^{-5} (moles)		

Total Question 8		16
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